

B.E / B.Tech. PRACTICAL END SEMESTER EXAMINATIONS, APRIL/MAY 2026

Fourth Semester

CS23421- Database Management Systems Laboratory

(Regulations 2023)

Time: 3 Hours

Answer any one Question

Max. Marks 100

Aim & Algorithm	Queries/ Program	Execution	Output	Record	Viva	Total
10	40	20	10	10	10	100
1. Design and implement a University Academic Database for student-course management. Normalize up to 3NF, define constraints, insert sample data, and develop PL/SQL modules for result processing and reporting.						
2. Given: STUDENT(SID, NAME, DEPT) MARKS(SID, SUBJECT, SCORE) Write SQL queries to find department-wise toppers, subject averages, and students scoring above overall average using joins and subqueries.						
3. Develop a Hospital Management System with patient, doctor, and appointment tables. Implement triggers for audit logging and procedures for billing, and generate analytical reports.						
4. Given: EMP(EID, NAME, DID, SALARY) DEPT(DID, DNAME) Write SQL queries to find highest paid employee per department, departments with average salary above a threshold, and demonstrate correlated subqueries.						
5. Design a Banking System supporting account and transaction management. Demonstrate TCL commands (COMMIT, ROLLBACK, SAVEPOINT) and implement a fund transfer procedure ensuring consistency.						
6. Given SALES(PID, CID, AMOUNT, DATE): Write queries to compute monthly revenue, identify top 5 customers, and calculate average sales using aggregation and grouping.						
7. Design a Library Management System. Implement fine calculation logic using functions, triggers for overdue books, and generate usage reports.						
8. Given ORDER(OID, CID, AMOUNT): Write SQL queries using nested queries (IN, EXISTS, ANY, ALL) to identify high-value customers and customers with no orders.						
9. Design an Online Retail Database. Implement discount calculation using functions, order processing using procedures, and perform sales analytics.						

10. Given STUDENT, COURSE, ENROLL tables: Demonstrate INNER, LEFT, RIGHT, and FULL JOIN operations and compare their outputs with proper examples.
11. Design an Employee Payroll System. Implement salary computation using functions and procedures and generate department-wise salary reports.
12. Given ACCOUNT(AID, BALANCE): Create triggers to prevent negative balance and maintain transaction history. Demonstrate trigger execution with examples.
13. Design an Airline Reservation System. Implement constraints, triggers for seat allocation, and queries for booking analysis.
14. Given CUSTOMER table with large data: Create views for restricted access, implement indexing, and analyze query performance improvement.
15. Design an Inventory Management System. Implement triggers for automatic stock updates and generate reorder-level reports.
16. Demonstrate TCL and DCL commands: Perform a transaction using COMMIT, ROLLBACK, SAVEPOINT and apply GRANT/REVOKE for different users with explanation.
17. Given the table: PRODUCT(PID, NAME, CATEGORY, PRICE, STOCK) Perform the following: Write queries to identify top-selling categories (assume sales logic if needed) Find products with low stock levels Create indexes on frequently queried columns Analyze and justify performance improvements
18. Design a NoSQL document-based schema (JSON) for customer orders. Perform retrieval operations and compare with relational schema.
19. Develop a Firebase Real-time Database application for task management. Demonstrate real-time updates and analyze consistency issues.
20. Given a retail dataset: Perform analytical queries to compute category-wise sales, identify trends, and interpret results for decision-making.

Internal Examiner

External Examiner